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Interacting and reflecting together in VR-mediated group learning: Immersive and observational experiences in shared mixed reality☆

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ABSTRACT

Immersive Virtual Reality (VR) is increasingly used in education to simulate real-world scenarios and support experiential learning. However, limited access to VR equipment remains a barrier to widespread adoption. This case study explores how VR can be effectively integrated into group learning by distributing roles among learners: a Performer, who interacts directly within the immersive VR environment, and Observers, who engage from outside the headset. Conducted in the context of a pedestrian safety training program, the study involved 45 second- and third-grade students in nine small groups. Through observation data analysis, the study identified distinct patterns in how Performers and Observers engaged in the VR simulation and reflection process. Performers exhibited real-time decision-making under pressure, while Observers demonstrated meta-cognitive awareness and collaborative sense-making. During the post-assessments, no statistically significant differences in safety knowledge or behavioral performance were found between Performers and Observers; the need for future studies with larger samples remained. The study identifies a four-stage guided reflection process that enabled groups to synthesize individual and shared experiences. Additionally, the study conceptualizes a Shared Mixed Reality learning space, where virtual and physical worlds are merged through peer collaboration and group reflection. These findings offer implications for future research and practice in VR-mediated group learning and reflection, highlighting VR's adaptability to resource-limited educational settings.

1. Introduction

Learning through concrete experiences is essential for students' meaning making and knowledge construction (Dewey, 1933; Kolb, 1984), particularly for children whose abstract reasoning is still developing (Piaget & Inhelder, 1972). Virtual Reality (VR) is emerging as a promising educational tool to provide such experiences by offering immersive and authentic environments that overcome barriers of distance, cost, safety, and accessibility (Radianti et al., 2020; Santilli et al., 2025). In particular, immersive VR enables students to navigate, manipulate virtual objects, and engage socially in three-dimensional digital worlds (Cummings & Bailenson, 2016; Shi et al.,

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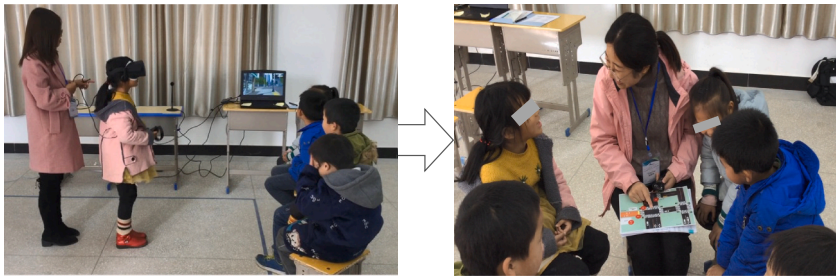

Fig. 2. *Two Group Activities: Group VRSimulation and Group Reflection.*

activities to capture students' perspectives on VR-mediated group learning, and researchers' field notes were collected as supplementary data. All video and interview recordings were transcribed verbatim.

Following the VR-mediated group learning, students completed a knowledge test and a VR-based behavior assessment. The knowledge test was a ten-item pen-and-paper measure evaluating students' understanding of traffic signals and decision-making in various situations. The behavioral assessment evaluated students' actions at 11 decision-making points in the VR program.

3.5. Observation data analysis

This study employed an inductive data analysis approach to analyze the nine group cases. As a bottom-up approach, the inductive data analysis process allowed the data to speak for itself through the researcher's in-depth investigation of the raw data (Bendassolli, 2013). There were no preconceived hypotheses or theories used in the analysis process. In general, data analysis progressed in four steps adjusted by Braun and Clarke's (2006) thematic analysis procedure, including generating initial codes, identifying pattern codes, identifying themes, and defining themes. The four steps merge related procedures from the original six-step guide and focused on the theme generation. A qualitative data analysis program, NVivo Version 12, was used to store, organize, and manage the data collected in this study.

3.5.1. Step 1. Generate initial codes

The first step involved identifying and labeling initial behavior codes from video data, organized by group case. Three guiding decisions shaped the coding process (1) Codes were categorized by role (e.g., *Performer*, *Observer*, and *Facilitator*): to support role-based analysis; (2) Coding units were short, distinct behavioral episodes due to the time-based nature of video data; and (3) Behaviors were coded separately for the group VR simulation and the group reflection, aligned with the research questions. An intercoder agreement procedure was used to ensure reliability of code generation (Lincoln & Guba, 1985), with the first author conducted initial coding and the second author independently coded two randomly selected cases. Initial percent agreement was 85%, which increased to 97–98% following discussion and consensus resolution. The first author then coded the remaining cases, with a final spot-check confirming 97% agreement. As a result, 103 behavior codes were identified from the group VR simulation phase (28 for Performers, 75 for Observers, see Appendix B), and 148 codes from group reflections phase (32 for Performers, 54 for Observers, 47 for the Facilitator, 15 for the group as a unit, see Appendix C).

3.5.2. Step 2. Identify pattern codes

In Step 2, the initial codes were aggregated into pattern codes through iterative discussions between the two coders (Bazeley, 2013). This step aimed to synthesize the data into meaningful categories that captured the roles' actions across the learning phases. In the VR-mediated group simulation phase, the Performer's behaviors were grouped into five categories, the Observer's behaviors fell into six (see Appendix B). Off-task behaviors were excluded to maintain focus on actions related to the research questions. During the group reflection phase, twelve pattern codes were identified, with Facilitator, Performer, and Observer behaviors each grouped into four categories (see Appendix C). The initial codes and pattern identification set the foundation for further analysis and theme generation (Yin, 2018).

3.5.3. Step 3. Identify and review themes

Guided by research questions, the researchers generate themes based on continuous discussion and negotiation in Step 3. Three themes emerged to represent the Performer's experience in VR-mediated group simulation, and six themes were identified to show the Observer's experience (RQ1). Four stages of group reflection emerged from the pattern codes, indicating how the Performer, Observer, and Facilitator collaboratively recall, analyze, and learn from the VR experiences (RQ2).

3.5.4. Step 4. Define themes and produce the report

The final step was defining and naming themes, which mainly focused on finalizing the themes and associated definitions. The four steps of thematic analysis, along with Appendices B and C, captured the progression from initial codes to patterns that informed the final theme generation. Based on continuous discussions and negotiations between researchers, this step yielded findings that

addressed the research questions.

3.6. Outcome data analysis

Two types of learning outcomes were analyzed (RQ3). The knowledge test consisted of ten multiple-choice items on safe pedestrian behavior. Each student's score was calculated by dividing the number of correct answers by the total number of questions, resulting in a score between 0 (0% accuracy) and 1 (100% accuracy). For the behavioral outcome, a safe pedestrian behavior rubric was applied to assess students' actions at the 11 decision-making points in VR. Scores were determined by dividing the number of safe actions by the total decision-making points, yielding a safety score ranging from 0 (0% safe decision) to 1 (100% safe decisions).

Two trained raters independently scored the recorded behaviors using the rubric. Interrater reliability was assessed using Cohen's kappa, indicating high agreement between raters ($k = 0.94$, $p < .001$). The few discrepant decision-making points were resolved through a second review and discussion until consensus was reached, and the agreed final pedestrian performance scores were used for data analysis.

Given the VR-mediated group learning setting, the four Observers within each group shared the same VR simulation and group reflection experience. Analyses were therefore conducted at the group level to preserve the independent assumption. The mean learning outcome scores of the four Observers were computed and paired with the corresponding Performer's scores, yielding nine matched pairs for analysis ($n = 9$). Wilcoxon signed-ranked tests with continuity correction were then used to compare the learning outcomes between the Performer and Observer roles. (Field, 2013). Data analysis was conducted using R (version 4.4.0; R Core Team, 2024) in the RStudio integrated development environment (version 2024.4.1.748; Posit Team, 2024).

4. Results

4.1. RQ1: Role-based experiences in VR-mediated group simulation: Performer versus observer

4.1.1. Performer in the virtual spotlight

In the VR-mediated group simulation, the Performer's behaviors were grouped into five categories: (a) Spontaneous Action, (b) Independent (Conscious) Action, (c) Responsive Action, (d) Influenced Action, and (e) Active Interaction with Others (see Appendix B). Three themes emerged from the pattern codes to capture the Performer's experience. Firstly, the Performer's role involved both benefits and drawbacks. Their Independent Actions (e.g., interact with objects in VR) indicated the Performer's access and control over the virtual environment, while certain Spontaneous and Responsive Actions revealed a loss of awareness of physical surroundings (e.g., lose balance). Meanwhile, the role of Performer carried implicit responsibilities. Their behaviors were largely driven by the goal of completing the VR tasks in the presence of others, whether through Spontaneous Actions (e.g., respond to the VR stimuli), Conscious Action (e.g., make decisions), or Active Interaction with Others (e.g., seek help). Furthermore, the Performer actions and decision-making in VR were affected by Observers in the group setting. Their behaviors were shaped both unconsciously (e.g., spontaneous reaction to directions) and consciously (e.g., actively ask for others' suggestion).

Empowerment. The Performer navigated a dynamic landscape of gains and losses inherent in immersive VR technology. They enjoyed full access to multimodal experiences and hold significant control over the virtual world. Accordingly, the Performer role empowered the student to make decisions in VR to shape the outcomes and directions of the virtual scenarios. However, this immersion came at the cost of disconnecting from the physical world. The Performer's actions in Group Case #3 demonstrated her autonomy in VR, while also revealing reduced awareness of her physical surroundings:

> … as the pedestrian traffic light turned to green in the virtual city, the Performer immediately initiated movement towards the crossroad. Stepping forward with purpose, she intent to cross the road within the simulated environment. In her attempt to navigate the virtual environment, she momentarily lost balance, prompting her to regain stability in the physical world … as she moved in the virtual city, physically she was moving toward the edge of the VR working station and almost hit a chair. The safety guide directed her into the center of the zone. (extraction and description of the Performer's actions from Group Case #3)

Responsibility. With the privilege of access and control in VR, the Performer role took unspoken responsibilities that required efforts and focus to fulfill. When a student volunteered or was assigned to this role, they had to wear the VR headset and use controllers to interact with the virtual world. The role carried to a tacit expectation: completing tasks in the VR program with their peers observing. This expectation led to a list of duties and requirements. The student needed to orient and adjust themself in the virtual environments, overcoming any difficulties caused by the setting (e.g., can't see one's physical environment and learning how to navigate the virtual space). To complete tasks in the virtual city, the Performer needed to apply certain pedestrian knowledge and make decisions in complex situations. Importantly, the Performer was not isolated in their actions; they were accompanied by a safety guide, four peer Observers from the same class, and researchers collecting data. The presence of these individuals could lead to unpredictable factors, such as pressure, distraction, or assistance, affecting the Performer's experience. For instance, in Group Case #1, one Performer struggled during the task:

> … suddenly, the Performer couldn't move forward in the virtual city. She tried stepping forward and waving her arms as directed, but she couldn't get into the crosswalk even though the traffic light was green. Her peers were chatting, and the safety guide tried to direct her somewhere she couldn't see in the physical world. The guide left to check the VR program, leaving her standing still, facing her peers with the headset on. The four Observers chatted and laughed and turned silent …

Influence. With peer Observers involved, the Performer's actions and decision-making were inevitably influenced. Since the learning group consisted of classmates, pre-existing social relationships could affect the group dynamic. The mere presence of Observers could lead to changes in the Performer's behavior, as they became more self-conscious and aware of being watched. This awareness can cause subtle shifts in the Performer's actions. Moreover, the Performer often actively seek input from Observers, inviting them to contribute to their decision-making process. For example, in Group Case #3, the Performer asked for confirmation from the Observers before acting, "can I go (cross the road) now?" Beyond invited involvement, Observers could directly influence the Performer's decisions by offering advice or sharing information. In Group Case #5, Observers guided the Performer, suggesting specific actions that reflected their own ideas and preferences, "go forward, you'll see a parking lot!" The presence of multiple Observers can also create additional layers of intervention, as their conversations or interactions may indirectly impact the Performer's choices. For instance, the Performer's decision-making can be distracted by the discussions happening between Observers, further complicating their experience in the simulation.

4.1.2. Observers in the virtual periphery

The Observer's behaviors fell into six categories: (a) Engaged Observation, (b) Spontaneous Action, (c) Responsive Action, (d) Proactive Action, (e) Interaction with Performer, and (f) Interaction with Other Observers (see Appendix B for details). Six themes emerged from the pattern codes to represent the Observer's experience. Specifically, the Observer obtained the sense of mental immersion through engaged observations (e.g., the Observer reacts spontaneously to the virtual stimuli). The Observer interacted with the virtual environment through Interaction with Performer. By helping the Performer complete the VR tasks, the Observer gained a sense of ownership in VR experience (e.g., think actively about solving problems in VR, prevent others from disturbing the Performer). Furthermore, the Observer possessed the flexibility to shift perspectives between the real and virtual worlds. With multiple observers in the group settings, the Observer played different roles and was influenced by the presence and actions of other Observers.

Engagement. The Observers were found highly engaged in the VR simulation even though they didn't have direct access to VR. The Observer obtained a sense of mental immersion in VR through engaged observation. Observers displayed spontaneous reactions to the virtual environment stimuli, using body language to simulate movements (e.g., Group Case #1, #6, the Observers stand up to step forward and wave hands, imitating the moving actions in VR). Additionally, they mentally placed themselves in the VR environment, often verbalizing thoughts as if they were the ones acting in VR. For example, one Observer remarked, "only if I can walk faster (in VR)" while watching the Performer cross a road (Group Case #2).

Influence. Another approach the Observer used to engage with the virtual world was by influencing the Performer's actions. Although they had no direct control over the VR program, the Observer shaped the experience by verbally guiding the Performer. They often provided real-time directions or suggestions to influence decision-making, actively participating in the task completion. For example, in Group Case #9, Observers urged the Performer to "run! Go faster, faster!" before the traffic light changed, showcasing how their input altered the course of the simulation. This indirect interaction allowed Observers to engage meaningfully with the VR content, despite not being physically immersed.

Role Shifting. Additionally, the Observer benefited from the ability to shift perspectives between the virtual and real worlds. Unlike the Performer who was fully immersed in VR, the Observer watched the simulation on a desktop while remaining aware of their physical surroundings. This awareness allowed them to discuss strategies with peers and plan their own actions in VR from outside the headset. By observing the Performer's actions and mentally engaging with the task, they could actively participate through indirect influence while imagining their future performance in similar situations. In Group Case #7, Observers demonstrated their engagement by combining mental immersion, influencing the Performer's actions, and shifting perspectives between the virtual and real worlds.

> … As the Performer approached the road, he paused to check both sides for approaching vehicles. Observing the VR display from the desktop, one Observer commented, "I would look up to check the traffic signal." In response, the Performer looked up, and the traffic light shown green, signaling it was safe to cross … (Researcher memo: this moment exemplifies how the Observer's input influenced the Performer's actions, bridging their own perspective with the virtual world and actively engaging in the simulation process).

Multiple Roles. Though all engaged in group VR simulation, the Observers were different. The analysis revealed that Observers played varied roles throughout the simulation. Some were passive bystanders, merely watching without much involvement (Group Case #4), while others imagined themselves in the simulation, mimicking actions as if they were co-players (Group Case #2). Certain Observers acted as directors, influencing the Performer's actions to reflect their preferences (Group Case #7), while others disrupted the process with distracting behaviors (Group Case #3). Many also took on the role of helpers, guiding the Performer to successfully complete the assignment (Group Case #9).

Shared Ownership. Notably, the Observer developed a sense of ownership by helping the Performer accomplish the assignment. Although there was no pre-determined rule regarding how the group worked together, most group cases showed collaboration in completing the simulation tasks. Specifically, the Observer supported the Performer in completing the assignment by providing suggestions/directions (e.g., Group Case #9), encouragement (e.g., Group Case #2), and efforts to prevent disruption and promote the process (e.g., Group Case #3). Finishing the VR simulation became not merely the Performer's mission but a shared and collaborative experience for the Observers.

Interaction Between Observers. Since there was more than one Observer in a group, Observers interacted with each other as well as with the Performer. Different interactive behaviors were found between Observers, suggesting the Observer's experience was also influenced by others who shared the same role in the group. For example, Observers were found to share their observed information and opinions with each other. Furthermore, when an Observer directs others' attention to their focus, he influences other Observers'

information-taking from their perspective. Meanwhile, Observers were also found to interact with each other during the simulation, including discussing the Performer's actions and exchanging opinions on the VR tasks. Additionally, Observers were found to encourage or prevent other Observers' actions in the group, leading to unique experiences created by the interactions.

4.1.3. Behavioral comparison of two roles in VR-mediated group simulation

Based on the identified pattern codes of the Performer's and Observer's actions (Appendix B), shared behavioral patterns across both roles were subsequently compared. Accordingly, five categories of behavioral patterns were identified to compare the two roles' actions in VR-mediated group simulation, including (a) Spontaneous Action, (b) Responsive Action, (c) Independent/Proactive Action, (d) Influenced Action, and (e) Interactive Action (see Table 1).

Although the Performer and Observer had different access to the VR program—one was fully immersed in VR with control, and the other saw the virtual world from projection—they both had active and responsive actions to the virtual world. The Performer actively engaged with and responded to virtual objects to accomplish tasks within the VR program, while the Observer reacted to the virtual environment indirectly, often directing the Performer to act as they themselves might in VR. This dynamic led Observers to partially perceive the Performer as their avatar, shaping the experience through guidance and feedback. Accordingly, both roles displayed spontaneous, responsive, active decision-making actions toward the tasks in VR. Meanwhile, interactive behaviors between the two roles also happened in VR-mediated group learning, where they were connected through the real-time projection of the VR program and free communication between each other. Both roles were aware of their role differences compared to others in the group, and they were motivated and cognitively engaged in problem-solving within the VR program. Additionally, both roles' behaviors and actions were influenced by others, resulting in a co-constructed and shared experience in the group rather than an independent one.

The difference between the two roles' behaviors was also apparent. Since the Performer was the only person simulated in VR, the role was exposed more to the virtual world, triggering various spontaneous actions during the simulation process. Meanwhile, since the Performer had the control and implicit duty to accomplish tasks in VR, the role enacted more conscious and independent actions than the Observer. Furthermore, the Performer was the only person in the group who could actually make an impact on the virtual world, therefore, endured others' presence and active involvement to enact influenced behaviors. In contrast, the Observer's behaviors were based upon their notice of the Performer's actions and the virtual world projected on the screen. The two roles had different goals when they joined the group with assigned roles and resources, and they used various means to enact their thoughts and impact the virtual environment.

4.2. RQ2: Group reflection on the VR-mediated group experience

The inductive data analysis identified four distinct stages of group reflection, indicating how the Performer and Observers reflected on their unique and shared experiences with the guidance of the Facilitator. The three roles shown specific behaviors during each stage. These patterns aligned closely across roles and were guided by the Facilitator's scaffolding. Final emerged themes with 12 role-specific actions (four core actions by three roles) is presented in Table 2 (see Appendix C for initial codes). Specifically, the first stage, Guided Retrospection and Description of Experience, focused on reviewing critical experiences for further analysis. The Performer and Observer shared their individual and collective experience, and the Facilitator provided guidance for the group to recall critical details and their thoughts during the experience. The second stage, Guided Analysis and Evaluation of Experience, suggested the group's deliberation on their VR experience. The Performer and Observer evaluated and analyzed their individual and shared VR experiences with the support of the Facilitator. The third stage, Guided Group Exploration and Discussion, indicated the group's guided teamwork in examining and reconsidering their VR simulation experience. The final stage is Learning and Conclusion, which indicated the group's meaning-making and future decisions from the group reflection. These stages demonstrate that the group, despite having different roles, engaged in a structured and collaborative process to reflect on both immersive and observational experiences. While these stages provide partial answers to RQ2, additional findings from the analysis offer valuable insights into the dynamics of group

Table 1
The Performer' and Observer's behaviors in VR-Mediated group simulation.

Behavior	*Performer*	*Observer*
Spontaneous	React to the VR environment	React to the VR environment
	React to the *Observers*' intervention	React to the *Performer*'s actions
	Navigate in the virtual environment	Imagine walking in VR
	Lost balance in the physical environment	
Responsive	Respond to the Guider's direction	Respond to the Guider's direction
	Respond to the *Observers*' comments/requests	Respond to the *Performer*'s needs/requests
		Respond to other *Observers*' actions
Independent/Proactive	Have control to act as one wishes	Influence the *Performer* to act
	Conscious decision-making in specific situations	Conscious decision-making in specific situations
	Actively interact with the virtual objects	Engaged observation
		Comment/think aloud regarding one's observation
Influenced	Act to get others' attention	Act to get others' attention
	Act according to others' direction/suggestion	Enact or stop certain actions due to others' encouragement or prevention
Interactive	Actively interact with *Observers*	Actively interact with the *Performer*
		Actively interact with *Observers*

Table 2
Twelve role-specific actions across four group reflection stages.

Group Reflection Stage Implemented	Core Action Emerged	Facilitator	Performer	Observer
Stage 1:	**Retrospection**	**Guidance on retrospection**	**Recall and describe one's experience**	**Recall and describe one's experience**
Guided Retrospection and Description of Experience	Review of critical experiences	• Helps the group to return to and discuss their VR experience. Students are guided to recall what they saw, heard, thought, and did in critical events that need further analysis. • Identifies critical information and leads for detailed description, e.g., "what did you notice when you crossed the street?" "I saw you stopped here, right? What happened?" or "I heard you say, 'run, there is a flashing light' what did you think at that moment?"	• Indicates and describes what they noticed, thought, and did during the VR simulation. The *Performer* also uses body language or gestures to represent what happened in the experience. For example, "I saw the pedestrian traffic light is red at that time.", "I was thinking I can't go (crossing the road)."	• Indicates and describes their observations and actions during the *Performer*'s VR experience, recalling what they noticed, thought, and did during the VR simulation. • Uses body language or gestures to represent what happened during the experience. • Describes and comments on the *Performer*'s action. For example, "I saw you (the *Performer*) dashed into the street.", "the traffic light was red for a while, but no car was on the road."
Stage 2:	**Analysis**	**Support on experience analysis and reconsideration**	**Analysis and (re)think about one's experience**	**Analysis and (re)think about one's experience**
Guided Analysis and Evaluation of Experience	Careful examination of experiences and consideration of potential solutions	• Asks individuals and the group to interpret critical signals/events, explain their opinions and actions, and make decisions in specific situations. • Asks questions to trigger students' analysis and reconsideration of their experience and observation. Questions include: "what does it mean when the traffic light flashes?", "Why did you act like that?" and "what if you rushed into the road at that moment?"	• Analyzes and thinks about the experience to review their knowledge and actions. The thinking process includes interpreting critical signals/ events, explaining their opinions and actions, and making decisions in specific situations. For example, "the flashing light means no time to cross the road" and "I would wait for another green light."	• Analyzes and thinks about their observations and experience to review their knowledge and construct meanings. The thinking process includes interpreting critical signals/events, explaining their opinions and actions, and making decisions in specific situations. For example, "the flashing green light means it will turn to red light soon," and "he was so close to the school bus. If the driver didn't see him, he would be hit to death."
Stage 3:	**Collaboration**	**Facilitation on teamwork/group discussion**	**Make meanings with others**	**Make meanings with others**
Guided Group Exploration and Discussion	Interaction and teamwork for the shared goal of learning	• Guides the group to participate in the discussion to share opinions and exchange ideas. • Leads the group to focus on the topic, creating a collaborative group atmosphere and inducing discussion with the involvement of group members. For example, the *Facilitator* directs the group's attention to the topic, "do you agree with that? (turns to two *Observers* who are not paying attention)" and "let's look at this first, okay?" The *Facilitator* invites everyone to participate in the discussion, "What do you think? (*Facilitator* turns to those who did not answer the question)", "what would you do if you were there?"	• Constructs meanings collaboratively from the analytical discussion with others. • Shares their opinions as well as responds to others' others' thoughts. The interactive meaning-making process includes expressing agreement/ disagreement, complimenting/ challenging others' perspectives, and responding to others' ideas with actions/illustrations. For example, "(responding to one *Observer's* comment, the *Performer* said) I would do the same."	• Constructs meanings collaboratively from the analytical discussion with others. • Shares their opinions as well as respond to others' others' thoughts. The interactive meaning-making process includes expressing agreement/ disagreement, complimenting/ challenging others' perspectives, and responding to others' ideas with actions/illustrations. For example, "I would not do like that (hurry up to cross the street). I would wait for another green light to cross the street."
Stage 4:	**Knowledge Construction**	**Support on knowledge construction and learning transfer**	**Construct understanding and personal decisions**	**Construct understanding and personal decisions**
Learning and Conclusion	New understanding and conceptual change	• Supports the group in understanding and coming up with safe pedestrian knowledge and behaviors in specific situations. • Helps the group construct meanings and decision-making regarding what they experienced in the VR program. For example, the *Facilitator* suggests and summarizes safe solutions in a specific condition, "so, we need to	• Concludes with safe pedestrian knowledge or decision-making in specific situations. • Acquires understanding and potential behaviors from experience, such as predicting consequences from hypothetical situations and making decisions in a given situation. • Links the VR experience to their real-life situations and generates their	• Concludes with safe pedestrian knowledge or decision-making in specific situations. • Acquires understanding and potential behaviors from experience, such as predicting consequences from hypothetical situations and making decisions in situations. • Links the VR experience to their real-life situations and generates their conclusions to other contexts

(*continued on next page*)

Table 2 (*continued*)

Group Reflection Stage Implemented	Core Action Emerged	Facilitator	Performer	Observer
		check both sides on the road to ensure the vehicle stops when we cross the street."	conclusions to other contexts and illustrates a better individual understanding. For example, "(the *Performer* compliments what the *Facilitator* mentioned) you still need to check on both sides of the road (beyond traffic light), if you don't check, there could be cars."	and illustrate a better individual understanding. For example, "(an Observer concluded) so you have to continue crossing the street in that situation (where the traffic light started flashing when you in the middle of the road)."

reflection.

4.2.1. Facilitator in group reflection

Essential Role. The importance of the Facilitator in guiding group reflection became evident, especially for students lacking experience in teamwork and reflective practices. The Facilitator led the reflection by asking prompt questions and providing feedback, ensuring students stayed focused on critical experiences. Additionally, the Facilitator shown three critical skills: knowledge of the subject matter being reflected upon, an understanding of reflective processes to foster meaningful discussions, and the ability to create a supportive environment for collaboration and teamwork. The direct and observational experiences were connected due to the Facilitator's deliberate guidance.

Challenging Position. Practical challenges were identified in the Facilitator's role. One challenge is that students may become overly dependent on the Facilitator's guidance, which can limit their autonomy and active participation in the reflection process. Additionally, the Facilitator's position of authority, particularly when there is power or knowledge imbalances, can affect group dynamics and hinder open dialogue. Lastly, managing group reflection, especially in larger groups, proved demanding. The Facilitator found it difficult to balance multiple students' inputs, handle conflicts, and maintain focus on various topics, as demonstrated by the memo.

> I think I was playing an essential role in the group reflection as the students relied on my guidance to review and analyze their experiences … it was stressful when five children in the group talked to me at the same time, especially when they were focusing on different things. They all wanted my attention, but I could only lead one topic at a time. I found myself forgetting to follow up with other topics after letting one child share their opinion first, let alone handling conflicts or disagreements between different individuals (Facilitator's memo and reflecting notes).

4.2.2. Performer in group reflection

Social Pressure. The group reflection was a different experience for the Performer compared to VR simulation. The Performer experienced social pressure due to being the focus of the reflection process, as their actions were central to the discussion. This spotlight could cause discomfort, especially when Observers were critical or judgmental. For example, in Group Case #2, the Observers critiqued the Performer's risky behavior in VR during their group debriefing, leading the Performer to withdraw emotionally and become less engaged in the discussion. The Facilitator had to intervene to reintegrate the Performer into the group dialogue. Additionally, Observers often overlooked the Performer's retrospection, interrupting or answering questions on their behalf, which diminished the Performer's opportunity to reflect openly.

Cognitive Load. The Performer faced a high cognitive load from being fully immersed in VR, which might hinder their ability to recall their experiences accurately for meaningful reflection during the discussion. Immersive environments provided complex, multimodal stimuli, making it difficult for the Performer to focus on key details during reflection. In Group Case #5, for instance, the Performer missed critical cues noticed by the Observers. Furthermore, Performers sometimes struggled to remember details from the VR experience, as seen when one Performer in Group Case #2 was unable to recall specific events, despite wanting to contribute to the discussion, "I think I remember it, but I just can't recall it at the moment." This highlights the cognitive demands placed on the Performer during both the VR simulation and the reflection process. A few extractions from the group reflection videos show the challenges the Performer faced:

> The Facilitator faced toward the Performer and asked, "what color was the traffic light when you crossed the road?" The Performer looked back to the Facilitator, lifting her face in thought but remaining silent. Before she could respond, another student, who had been an Observer during the VR simulation, interjected, "It (the traffic light) was flashing before you arrived at the corner." The Facilitator redirected the question to the Performer, "Do you remember that?" The Performer, lowering her head and voice, quietly replied, "No." (Group Case # 5)

> … to answer the question "what did you see on the first street you cross?", the Performer replied, "a school bus." an Observer quickly corrected her, saying, "That happened on the third road, not at the beginning." (Group Case #2).

4.2.3. Observers in group reflection

Ownership. The Observer got a unique position in group reflection. They shared a sense of ownership over the VR simulation experience, though they were not the one simulating. They often interrupt the Performer to answer questions or share details from their own perspective. For example, in Group Case #6, an Observer used first-person language to describe the Performer's actions, saying, "I didn't see the traffic light, so I thought it was okay to cross." Observers also fill in details the Performer missed, such as when one Observer noted, "It was flashing before you arrived at the corner" (Group Case #5). Additionally, they challenge the Performer's recollections, as seen when an Observer corrected the Performer's memory of a school bus in Group Case #2.

Role Awareness. Even though showing ownership on the VR experience, the Observer remained aware of the differences between their role and the Performer's. Observers often compared themselves to the Performer, judging their actions and offering alternative decisions. For example, when the Performer stated that he would cross the road when the vehicles had stopped, the Observer disagreed, stating, "I would not do that. It is still risky. I would check the traffic light and vehicles." (Group Case #4). While discussing the Performer's behaviors, the Observer planned for their own VR experience. For instance, when the group debated whether the Performer's action was safe in VR, one Observer looked around and said aloud, "I will do something like this …" (Group Case #2).

Social Dynamics. Notably, there was a lack of social rules governing how Observers interact with others in the group. They frequently interrupted both the Performer and other participants to share their perspectives, often without realizing the impact. For example, in Group Case #1, when the Facilitator asked the Performer about her actions in a particular situation, the Observer interrupted and said, "She was waiting there because the traffic light showed red." Observers also struggled to handle differing opinions, sometimes causing tension, as in Group Case #2 when an Observer's comments upset the Performer. In some cases, Observers even retracted their suggestions after being challenged by peers (Group Case #3). For example, one Observer explained their thought process for a specific situation, stating, "I would cross the road when the flashing light comes on." Another Observer disagreed, questioning, "Who said that?" In response, the first Observer denied their statement, claiming, "I didn't say that."

4.3. RQ3: Learning outcomes from the VR-mediated group learning

4.3.1. The Performers' and observers' knowledge test

Both the Performer and Observer groups completed a knowledge test following the VR-mediated group learning to examine learning outcome related to pedestrian knowledge acquisition. Given the design of the group learning, four Observers within each group case shared the same VR simulation and group reflection experience. Therefore, analyses were conducted at the group level to preserve the independent assumption. For each of the nine group cases, the mean knowledge test score of the four Observers was computed and paired with the corresponding Performer's score, yielding nine matched pairs for analysis ($n = 9$). A Wilcoxon signed-ranked test with continuity correction was used to examine the difference between the Performer and Observer groups' learning outcome, considering the small sample size limits the reliability of the normality assumption required for parametric testing (Field, 2013).

The Wilcoxon signed-ranked test revealed no statistically significant difference in knowledge test between the Performer ($M = 0.86$, $SD = 0.13$) and Observer groups ($M = 0.75$, $SD = 0.23$), $V = 30.5$, $p = .093$, pseudomedian = 0.125, 95% CI [−0.050, 0.288]. The positive pseudomedian, the direction of the V statistic, and a confidence interval leaning predominantly positive collectively suggested a pattern in which Performers tended to score higher than the mean of their groups' Observers across the nine groups. Although the difference did not reach statistical significance at $\alpha = .05$, a large effect size was observed ($r = 0.574$, 95% CI [0.08, 0.89]). This suggest that the non-significant p-value is likely attributable to insufficient statistical power given the small sample of nine paired groups (Cumming, 2014). These findings indicate that while no statistically significant differences in knowledge acquisition were found in this study, a potential advantage for Performers that requires further investigation with larger samples.

4.3.2. The Performers' and Observers' behavior performance in VR

Both the Performer and Observer groups completed a VR-based behavior performance assessment following VR-mediated group learning to examine their application of pedestrian safety behaviors. As analyses were conducted at the group level, the mean behavior performance score of the four Observers within each group was computed and paired with the corresponding Performer's score, yielding nine matched pairs ($n = 9$). A Wilcoxon signed-ranked test with continuity correction was used to examine the difference between the Performer and Observer groups' behavior performance in VR after the group learning (Field, 2013).

The Wilcoxon signed-rank test revealed no statistically significant difference in behavior performance between the Performer ($M = 0.73$, $SD = 0.22$) and Observer ($M = 0.77$, $SD = 0.11$) groups, $V = 15$, $p = .726$, pseudomedian = −0.023, 95% CI [−0.223, 0.125]. The near-zero pseudomedian and a confidence interval symmetrically crossing zero indicated no consistent directional pattern observed across the nine groups, suggesting that Performers and Observers did not demonstrate statistically significant difference in pedestrian behavior assessment. Meanwhile, a small effect size was observed ($r = 0.119$, 95% CI [0.00, 0.72]), indicating a negligible difference between the two roles in behavior performance.

While the results indicated no statistically significant differences in behavior performance between Performer and Observer groups, a non-significant p-value does not provide evidence that the two groups are equivalent (Lakens, 2017). The confidence interval crossing zero, combined with the small sample size, indicates that the current data cannot definitively determine whether the groups are truly equivalent or meaningfully different. However, the small effect size suggests that any practical difference, if present, is likely minimal.

4.3.3. The Performer' and Observer's satisfaction and perceived learning experience

The closing interviews with the forty-five students provided valuable insights into their satisfaction and perceived learning experiences during the VR-mediated group learning activities. Overall, both Performers and Observers expressed positive feedback, enjoying the VR experience and recognizing its relevance to real-life situations.

For Performers, six out of nine preferred simulating the VR program with peers present, citing the support and suggestions they received during the activity. As one Performer shared, "I preferred group experience when others were around. They (the Observers) could give me suggestions like when I didn't notice the traffic light, they told me to go" (Group Case #3). In contrast, three Performers expressed a preference for experiencing the VR alone, mainly due to concerns about peer interruption and discomfort of being observed by others. For example, one Performer mentioned, "I prefer experiencing the VR by myself. They would laugh at me. I don't like that" (Group Case #4). However, all Performers reported enjoying the group reflection, valuing the opportunity to discuss the VR experience with their peers. Many highlighted the benefits of peer interaction, both in terms of fun and learning. One Performer noted, "I like discussing the experience with others because they would help me correct my mistakes (risky behaviors). I like to reflect with others because they would also learn things" (Group Case #2).

Observers reported similar experiences to Performers, with twenty-four indicating that they preferred engaging with the VR program themselves rather than observing. One Observer explained, "I prefer playing the program by myself than observing. It will be more fun. I like the scenes (in VR). I can do whatever I want" (Group Case #9). Five Observers preferred watching others due to personal discomfort with VR, citing reasons such as dizziness or fear. For example, one Observer shared, "I dare not play the VR program. I am afraid of being hit (by cars in the virtual world)" (Group Case #4).

5. Discussion and conclusion

This study investigated how learners engage, reflect, and construct knowledge in VR-mediated group learning through distinct roles: the Performer immersed in simulation and the Observer interacting from the periphery. By analyzing nine group cases, we explored the dynamics between these roles during simulation and reflection. The findings offered insights into how different forms of participation shape learning experiences in immersive environments.

5.1. The discovery of immersive VR-mediated group learning space

The immersive VR-mediated learning groups were formed by two roles with different resources and constraints. The Performers' and Observers' experiences in such a group setting can be explained using a learning space combining different types of realities (see Fig. 3). The blue area (A) represents the exclusive Virtual Reality, where Performers navigate without any physical world influence. In the nine group cases, Performers navigated in exclusive Virtual Reality when they received no intervention from their physical surroundings. For example, their Spontaneous and Responsive Actions to the virtual environment. In contrast, the exclusive Physical Reality indicates the space Observers positioned without knowledge of the virtual environment (the yellow area, C). The Observers' Proactive, Responsive, and Interactive behaviors occurred in physical reality with others from the physical world.

Importantly, the group setting created a novel learning space where the boundaries between the virtual and physical worlds blurred by a projected display. Observers in the physical environment were able to watch and understand what Performers were experiencing in VR. Thus, students from virtual and physical realities were able to communicate and collaborate. We define the overlapped space as

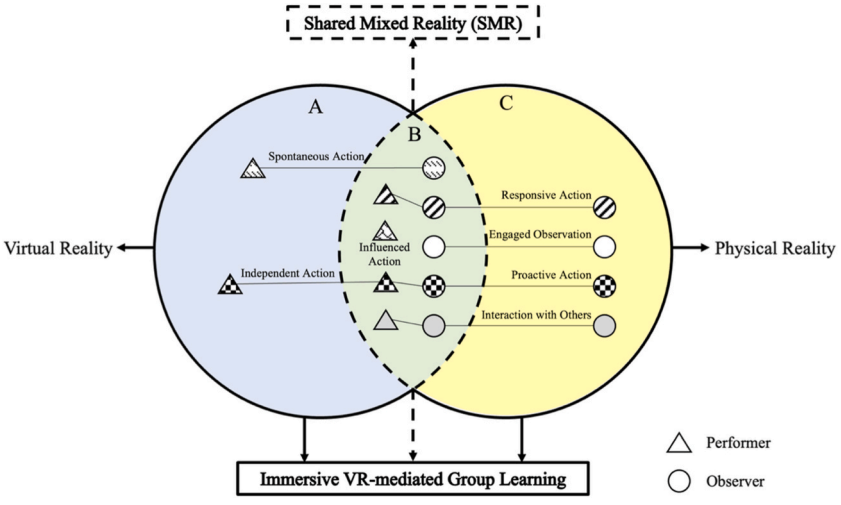

Fig. 3. *Immersive VR-mediated Group Learning Space Combining Different Spaces*
Notes. The triangle represents the Performer's actions, and the circle represents Observer's actions. Different fill patterns denote distinct types of behaviors, and the connecting lines between shapes indicate that they are the same type of behavior.

Shared Mixed Reality (SMR) in this study (the green area, B). For example, in Group Case #9, the Performer completed VR tasks in the presence of Observers. In SMR, the Performer was immersed in the virtual world while physically sharing the same space with peers. The Observers watched the Performer's actions and corresponding consequences in VR via a projected display. When the traffic light in the virtual scene turned red, the Observers called out, "Run! Go faster, faster!", prompting the Performer to respond accordingly. In this interaction, the Performer (re)acted consciously and unconsciously to others from their physical surroundings. As a results, the Observers impacted the virtual space by actively influencing the Performer's actions in VR. Thus, the SMR space provided a chance for those who were not immersed in VR to enact their thoughts in the virtual world. Accordingly, dynamic group interactions occurred due to the co-existence and overlapping of the virtual and physical spaces.

The discovery of immersive VR-mediated learning space offers several implications for future research. First, the study identified SMR as a shared space where learners exchange their physical and virtual experiences to achieve mutual awareness and interaction. Such a definition aligns with Benford et al.'s (1998) suggestion of Mixed Reality (MR), defining it as a shared space based on constructing and integrating transparent boundaries between real and virtual spaces. However, as Speicher et al. (2019) found from their interview with experts and literature review, the definition of MR is highly fragmented and diverse. Multiple terms are found to be used interchangeably with MR (e.g., augmented reality, virtual reality), and the MR experience is not clearly described and defined in studies. Accordingly, there is a need for further research to better define and characterize the notion of MR experiences. For example, unlike forms of MR that blend physical and virtual worlds for individuals, SMR is a specific form of MR that emphasizes interaction and mutual awareness between two or more individuals who have similar yet distinct experiences from virtual and physical realities. Second, the study highlights the potential of collaborative learning in VR-mediated environments, where one student engages in VR while others observe. This setup offers effective and interactive learning opportunities despite the limited availability of VR technology. The finding of SMR contributes to the interactions and exchanges of experience across physical and virtual experiences. Further research is needed to explore the nature of collaboration within these environments, particularly the different layers of group and sub-group interactions.

The VR-mediated group learning space also has some implications for instructional design. Specifically, the study revealed the importance of authentic and engaged learning situations, rather than the direct use of VR technology, in triggering mental immersion and cognitive engagement. Therefore, schools with limited resources of immersive VR equipment and resources can organize group learning activities to benefit more students with the observational learning mode. Additionally, the practice of VR-mediated group learning requires further considerations of effective scaffolding. Due to the division of labor and combination of multiple realities in VR-mediated group learning, students in such a learning environment need support to overcome challenges and collaborate with others. For example, instead of focusing solely on virtual or physical experiences, educators and instructional designers could intentionally design SMR experiences to connect and support learners in resource-demanding environments. This study examined the intervention of guided group reflection on students' interactions and learning outcomes, while more instructional strategies need to be designed and examined to support collaborative learning in VR-mediated group settings.

5.2. *The four stages of group reflection*

The study demonstrates four stages of group reflection, including (guided) retrospection and description of the experience, (guided) analysis and evaluation of the experience, (guided) group exploration and discussion, and (guided) learning and conclusion. Noticeably, the four stages are not sequential but interrelated. Fig. 4 suggests the relationship between the group reflection stages found in this study. Since group reflection is an activity accomplished by a group of learners rather than an individual, group interactions and communications play a central role in the process. Therefore (Guided) Group Exploration and Discussion (e.g., Facilitator invites Performer or Observer to share, discuss, or debate on their opinions), co-exists with and influences the other three stages of group reflection. The (Guided) Retrospection and Description of the experience (e.g., Facilitator asks Performer or Observer to recall what they saw/heard/thought) is the first step in the group reflection process, which allows the group to share their individual and shared impressions on the experience. The (Guided) Analysis and Evaluation of the experience (e.g., Facilitator prompts Performer or Observer to interpret/analyze what they saw/heard/thought) is a critical stage in the group reflection process, helping the group examine and reconsider their knowledge and behaviors during the experience. Finally, the (Guided) Learning and Conclusion (e.g., Facilitator supports Performer or Observer to summarize/conclude what they learn) is the last stage in the process, which suggests the results of group reflection, such as new understanding and plan for future behaviors.

Notably, the finding of the four stages of group reflection provides experimental support for the statement that reflection proceeds with systematic phases rather than a causal thinking process (Yang & Choi, 2023). Reflection has been disentangled into multiple stages or phases by a number of scholars (Atkins & Murphy, 1993; Bain et al., 2002; Boud et al., 1985; Boyd & Fales, 1983; Dewey, 1933; Gibbs, 1988; Korthagen & Vasalos, 2005; Rodgers, 2002). The group reflection stages found in this study validate the general

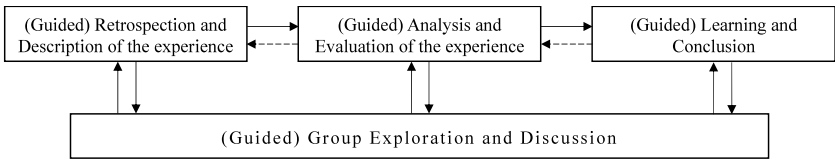

Fig. 4. *The Relationship between the Four Stages of Group Reflection.*

process of reflection while highlighting the unique characteristics of the group environment. Regardless of the social environment, the general reflection process includes retrospection, re-evaluation, critical analysis, and reorientation for thinking and action (Finlay, 2008; Koole et al., 2011). The group reflection has one more stage of *(Guided) Group Exploration and Discussion*, which reveals the unique characteristics of the group environment while adding to the complexity of the reflection process.

The findings of the four group reflection stages in this study have some implications for group reflection research and practice. Specifically, the four stages of (guided) group reflection require further investigation and validation in different learning contexts. Findings revealed in this specific study need further investigation in different contexts for generalization. This study also confirmed a couple of theoretical justifications for group reflection. For instance, the findings revealed that learners had varying focal points during their VR simulation experiences, which facilitated analysis on collective experiences. Moreover, the study demonstrated that individuals could identify their misconceptions and overcome limitations through group reflection. These findings align with the theoretical perspectives that suggest group environments offer opportunities for reflection and can assist individuals in achieving what they may not be able to accomplish alone (Eshuis et al., 2019; Johnson & Johnson, 1999; Vygotsky, 1978).

The findings from the group reflection stage also have implications for instructional design and reflective practice. The study confirmed the need for scaffolding to support group reflection, particularly for young learners who may lack collaboration and reflection skills. Future practices should focus on strategies to guide different stages of group reflection. Additionally, the stages of group reflection emphasized the importance of Guided Group Exploration and Collaboration, suggesting that teachers should prioritize practical facilitation strategies. Challenges such as a lack of shared understanding or social rules can hinder reflection, so educators could create strategies to foster open, collaborative environments that support meaningful group reflection.

5.3. Similar experiences and learning outcomes obtained by different roles

The study found that learners achieved similar experiences and learning outcomes despite their different access to VR. Both Performers and Observers were mentally immersed, cognitively engaged, and demonstrated motivation, agency, and ownership in problem-solving. Despite their role differences, both groups were aware of the team dynamic, actively interacting with peers and influenced by each other's presence. This suggests that VR-mediated group learning, where peers observe one student's VR simulation, can provides immersive and engaging experiences for all participants regardless of role. Post-activity assessments indicated no statistically significant differences in safety knowledge or behavioral performance between Performers and Observers, although the large effect size observed for the knowledge test ($r = 0.574$, $p = .093$) suggests the study was likely underpowered to detect a meaningful difference in knowledge acquisition. Nonetheless, these findings are broadly consistent with social cognitive theory (Bandura, 1986), which posits that observational learning can be as effective as direct experiential learning when supported by guided reflection. Rather than claiming equivalence of learning outcomes, the current findings suggest that the Observer did not demonstrably disadvantage the Performer under this design. This finding, however, requires cautious interpretation given the small sample of nine groups (Lakens, 2017).

Two key factors contributed to these similar learning outcomes. Firstly, the SMR space connected the Performer in the virtual world with Observers in the physical space, fostering collaboration and shared experiences (Benford et al., 1998). This connection encouraged both Performer and Observer to demonstrate high levels of ownership and engagement in VR simulation. However, distinct challenges emerged in this process: Observers couldn't directly enact their thoughts in VR, while Performers sometimes struggled with the disconnect between virtual and real worlds. The second essential factor was the guided reflection process. Guided reflection enabled both roles, whether through direct interaction with VR or observation, to learn from their experiences and integrate their insights into shared understandings (Yang & Choi, 2023). These two factors are interconnected in contributing to comparable learning experiences for Performers and Observers. SMR promotes shared experiences and interactions, which becomes foundations for meaningful group reflection. Guided reflection, in turn, helps learners analyze and transfer their shared experiences into co-constructed understanding. Challenges also emerged in the group reflection process, as Performers faced social pressure from peer judgment, while Observers found it challenging to navigate differing opinions. Addressing these role-specific challenges with tailored support, such as structured facilitation strategies for Observers and social-emotional scaffolding for Performers, may further enhance the effectiveness of VR-mediated group learning.

5.4. Implication

The findings of this study have some implications for research on VR-mediated group learning. The study supports the notion that observational experiences can be as effective as immersive ones, but success depends on well-designed instructional strategies, such as creating Shared Mixed Reality spaces and incorporating guided group reflection. Additionally, the study revealed distinct challenges associated with role assignments, such as Performers' reduced awareness of the physical environment and Observers' limited engagement in the virtual world. Addressing these challenges with targeted support can improve learning outcomes. Further research could explore scaffolding techniques to enhance collaborative learning, particularly in settings with limited access to technology.

The findings of this study also have practical implications for designing VR-mediated group learning. Designing group learning environments with limited VR resources can help address programmatic issues, such as equipment shortages, limited time for class activities, and teacher overloads, enabling schools and organizations with fewer resources to serve more students. However, instructional design is crucial to ensure that learners with varying access to resources achieve similar outcomes. The study shows that observational learning in shared physical and virtual spaces, combined with group reflection, can be effective, although a few challenges remain. For example, different roles and divisions of labor in VR-mediated group learning present unique challenges. The

findings indicate that Performers faced social pressure during group reflection, which can hinder their participation. Facilitators should be attentive to such challenges and create a supportive environment that *fosters collaboration and equitable learning experiences for all participants.*

5.5. Limitations and future directions

This study focused on a specific VR-mediated group learning context, examining how learners experience, reflect, and construct knowledge through two distinct roles—Performer and Observer. While the findings contribute to the theoretical understanding and practical design of VR-mediated group learning, the scope was limited to nine group cases. Although the current study examined the learning outcomes of the two roles (RQ3), the small sample size limits the generalizability of the findings and statistical power to detect differences. For example, while no statistically significant differences were observed between Performers and Observers in knowledge acquisition and VR behavior performance, a large effect size for knowledge ($r = 0.574, p = .093$) and a small effect size for performance ($r = 0.119$, $p = .726$) suggested uncertainty in the estimates. Larger sample sizes in future research would allow for more precise evaluation of potential role-specific differences.

Another limitation concerns the procedure used to assign students to the Performer and Observer roles. In this study, role assignment was based on voluntary selection, with random assignment employed in cases where no student volunteered. This approach introduces a potential self-selection bias, as students who volunteered for the Performer role may have differed from their Observer peers in characteristics that might be relevant to the study outcomes. For example, volunteers may have possessed greater technological familiarity with immersive VR environments, higher motivation toward the learning tasks, or greater willingness to perform in front of peers, which could influence how they engage and interact with others in the VR-mediated group learning setting. However, this cannot be confirmed without baseline measures of factors such as VR experience, motivation, or social confidence. Future studies could employ fully randomized role assignment to eliminate self-selection bias and strengthen causal inference regarding the differential effects of the Performer and Observer roles in VR-mediated group learning.

In this study, VR-mediated group learning was designed to integrate a shared mixed-reality space with guided group reflection. Although these components can be described as two phases, they are inherently interconnected, with experiences in simulation potentially shaping the quality and depth of subsequent reflection. Therefore, students' experiences in group VR exposure and subsequent group reflection jointly contributed to their learning outcomes. Future studies can examine the two group learning phases separately, as well as in combination, to further examine their individual and combined contributions to learning. Meanwhile, future research could further explore how these phases interact over time to influence both individual and collective learning. Investigating this connection can provide a more integrated understanding of how immersive VR experiences, when paired with structured reflection, support meaningful learning in group settings. For example, future studies could investigate how factors such as group cohesion, peer relationships, and role distribution influence learning processes and outcomes. Finally, the study's focus on young learners in a Chinese cultural context highlights the need for further research across diverse learning environments and cultural settings to ensure generalizability.

CRediT authorship contribution statement

Tingting Yang: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ikseon Choi:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Conceptualization. **Heng Luo:** Writing – review & editing, Validation, Supervision, Project administration, Investigation, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used ChatGPT (OpenAI) and Claude (Anthropic) to refine language, improve clarity, and ensure grammatical accuracy in selected sections of the manuscript. After using these tools, the authors reviewed and edited all content as needed and take full responsibility for the content of the published article.

Appendix A. Overview of the Group Cases and Participants

#	Grade	Members	Selection of Roles	A Glimpse of the Group
1	2nd	One girl and four boys	The girl was randomly assigned as *Performer*, as no one volunteered	VR simulation (8:29 min): the *Performer* acted quietly in VR, and *Observers* were engaged by interacting with each other but not with the *Performer*. Group reflection (8:37 min): active discussions guided by the *Facilitator*. The *Performer* and *Observers* share experiences and opinions based on the prompt questions.
2	2nd	Two girls and three boys	One girl volunteered to be the *Performer*.	VR simulation (4:06 min): active interactions between the two roles, with the *Performer* sharing and asking for helps and the *Observers* responding and directing the *Performer*

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#	Grade	Members	Selection of Roles	A Glimpse of the Group
			The other girl had VR experience before.	Group reflection (16:56 min): active discussions guided by the *Facilitator*. *Observers* interrupt the *Performer* to share their observations and opinions actively.
3	2nd	One girl and four boys	One boy volunteered to be the *Performer*.	VR simulation (5:18 min): the *Performer* attempted to interact with *Observers* while he acted to finish VR tasks, while *Observers* interacted with each other but not with the *Performer* Group reflection (9:31 min): active discussion guided by the *Facilitator*. The *Performer* and *Observers* talk through their experiences and opinions based on the prompt questions.
4	2nd	Four girls and one boy	One girl volunteered to be the *Performer*; she had VR experience before.	VR simulation (3:41 min): the *Performer* acted quietly and confidently, and others engaged in observing with few interactions Group reflection (9:55 min): discussion led by the *Facilitator*, with the *Performer* sharing actively and *Observers* listening most of the time
5	3rd	Four girls and one boy	One girl was randomly assigned as *Performer*, as no one volunteered	VR simulation (3:33 min): few interactions in the group with the *Performer* quietly acting and others engaging in observing Group reflection (7:15 min): discussion led by the *Facilitator*. The group responded to questions and prompts but not actively sharing opinions.
6	3rd	One girl and four boys	One boy volunteered to be the *Performer*; he had VR experience before	VR simulation (4:10 min): the *Performer* talked throughout the VR simulation, and the *Observers* engaged in watching with few interactions with others. Group reflection (7:51 min): active discussion guided by the *Facilitator*, with the *Performer* actively sharing and others passively responding.
7	3rd	Two girls and three boys	One boy was randomly assigned as *Performer*, as no one volunteered	VR simulation (3:33 min): few interactions during the VR simulation, with the *Performer* acting quietly and others engaged in observing Group reflection (8:48 min): active discussions guided by the *Facilitator*. The *Performer* and *Observers* share experiences and opinions together, with few interruptions to others' sharing
8	3rd	Two girls and three boys	One girl was randomly assigned as *Performer*, as no one volunteered	VR simulation (4:11 min): few interactions during the VR simulation, with the *Performer* acting quietly and others engaged in observing Group reflection (15:34 min): active discussions guided by the *Facilitator*. The *Performer* and *Observers* share experiences and opinions together, with few interruptions to others' sharing
9	3rd	Three girls and two boys	One boy was randomly assigned as *Performer*, as no one volunteered	VR simulation (3:51 min): the *Performer* searched for help and acted according to others' directions. *Observers* actively interacted with others. Group reflection (10:34 min): active discussions guided by the *Facilitator*. The *Performer* remains relatively quiet while the *Observers* answer questions and share their opinions on behalf of the *Performer*.

Appendix B. Results of Open Codes and Category for the VR Simulation Phase

Category | Open Code	Frequency	
	Case	Code
Role: Performer (5 categories excluding off-task actions, 28 codes)		
1. Spontaneous action/reaction in VR	**9**	**214**
Chuckle while performing in VR	1	2
Think out loud with their thoughts in VR	2	23
Have control in VR to act as they want	6	15
Have control in VR to obtain balance on their own	3	5
Have control in moving freely in VR	9	157
Lose balance with a certain action in VR	1	3
Unaware of the physical environment	4	9
2. Independent (conscious) action to accomplish the assignment	**9**	**199**
Actively interact with virtual objects	2	4
Freely explore the VR environment (not related to accomplishing the assignment)	1	4
Navigate in the virtual environment	9	97
Interact with the virtual environment to accomplish the assignment	9	69
Make safe decisions to complete the assignment	9	19
Not follow the *Observers*' direction (make their own decisions)	2	4
Not follow the VR guidance (make his own decision)	1	1
The *Performer*'s previous experience hinders his action in VR	1	1
3. Responsive actions in VR	**9**	**27**
Respond to the Guider's facilitation (reply to the Guider's question)	6	19
Look around to check objects mentioned by *Observers*	2	3
Reply to the *Observers*' comments	3	5
4. Influenced actions in VR	**9**	**39**
Act according to the *Observers*' request	3	3
Act with the information provided by *Observers*	2	2
Follow the Guider's direction	9	25

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Category | Open Code	Frequency	
	Case	Code
Follow the *Observers*' direction	4	9
5. Active interaction with others	**6**	**18**
Ask for confirmation of her desired action	2	2
Seek out help	4	4
Share her interactions with the virtual environment	2	5
Share out loud what they see in VR	3	7
Others (Off-task)	**2**	**7**
Ask questions about the VR program	2	4
Comment on the VR program	1	3
Role: Observer (6 categories excluding interrelated and off-task actions, 75 codes)		
1. Engaged (conscious) observation	**9**	**336**
Look at the *Performer*	9	79
Look at the screen	9	174
Look at the *Performer* and screen	9	65
Observers have different focuses for their observation	5	18
2. Spontaneous Action/Reaction during Observation	**9**	**80**
Imagine performing (mimic walking) in VR	7	18
Point to the *Performer*	3	11
Point to the screen	6	24
Think aloud, imagining he has control in VR	1	1
Think aloud, interpreting the virtual object	1	1
Think aloud, preparing for his own VR experience	1	1
Think aloud regarding the virtual scene	2	2
Think aloud as they are in VR	3	13
Use body language (gesture) to indicate the direction in VR	5	7
Use gestures to illustrate something to another *Observer*	2	2
3. Responsive Action/Reaction during Observation	**5**	**43**
Laugh out loud with the *Performer*'s response to their direction	2	4
Laugh after the *Performer* seeks help (with no action)	2	2
Look at the *Observer* who directs the *Performer*	2	2
Look at the *Performer* after the *Performer* makes certain actions in VR	3	13
Look at the *Performer* after the *Performer* seeks help	1	1
Reply to the *Performer*'s question	3	7
Respond to other *Observers*' suggestions on the virtual scene	2	4
Respond to the Guider's facilitation (on behalf of the *Performer*)	3	6
Respond to the *Performer*'s share with the virtual scene	1	1
Respond to the virtual scene with physical actions	2	3
4. Proactive Actions	**5**	**37**
Comment on the *Performer*'s action in VR	4	7
Comment on the task in VR (think like a *Performer*)	1	4
Describe the virtual scene	5	20
Predict/guess what happens next in VR	1	4
Predict/guess the *Performer*'s following action in VR	1	2
5. Active Interactions with the *Performer*	**6**	**65**
Ask the *Performer* about their virtual experience	1	1
Direct the *Performer* not to act	3	5
Direct the *Performer* to act	6	39
Encourage the *Performer*	2	2
Provide information to the *Performer*	5	10
Share strategies with the *Performer*	1	1
Suggest the *Performer* how to act	2	3
Talk with the *Performer* regarding the virtual scene		
Warn the *Performer* of dangerous situations or behavior	3	4
6. Active Interactions with other *Observers*	**6**	**45**
Challenge other *Observers*' direction on the *Performer*'s action	2	3
Discuss the *Performer*'s action in VR with other *Observers*	1	2
Discuss the virtual scene with other *Observers*	2	4
Illustrate something to another *Observer* with a gesture	1	1
Look at another *Observer* after observing something	2	9
Look at another *Observer* after the *Performer* makes a decision in VR	1	3
Prevent others from disturbing the *Performer*	1	2
Whisper to another *Observer*	7	21
Simultaneous and Interrelated Actions	**7**	**90**
Chuckle while interacting with other *Observers*	1	1
Chuckle while observing the *Performer*	5	14
Chuckle while observing the screen	3	3
Look at the *Performer* and laugh out loud	2	2
Look at the *Performer* and predict the following VR scene	1	1

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Category | Open Code	Frequency	
	Case	Code
Look at the *Performer* and smile	1	2
Look at the screen and comment on the *Performer*'s action	1	1
Look at the screen and describe the virtual scene	3	9
Look at the screen and direct the *Performer* to act	4	13
Look at the screen and discuss with other *Observers*	4	7
Look at the screen and imagine performing in VR	3	5
Look at the screen and predict the *Performer*'s action in VR	1	2
Look at the screen and provide information to the *Performer*	2	3
Look at the screen and reply to the *Performer*'s question	1	1
Look at the screen and suggest to the *Performer* how to act	1	1
Look at the screen and talk with the *Performer* regarding the virtual scene	1	1
Look at the screen and think aloud about the virtual scene	2	2
Look at the screen and warn the *Performer* of a dangerous situation	3	3
Look at the screen and whisper to another *Observer*	2	4
Talk to another *Observer* while mimicking performing in VR	1	1
Point to the *Performer* and whisper to another *Observer*	1	1
Point to the screen and describe the virtual event	5	6
Point to the screen and direct the *Performer* to act	1	1
Point to the screen and discuss with other *Observers*	2	2
Point to the screen and indicate the virtual object	1	2
Point to the screen and use gestures to indicate a direction	2	2
Others (Off-task)	**3**	**6**
Off-task attention	2	2
Off-task conversation between *Observers*	1	3
Off-task interaction between *Observers*	1	1

Appendix C. Results of Open Codes and Category for the Group Reflection Phase

Category|Open code, frequency	Frequency	
	Case	Code
Role: *Facilitator* (*4 categories, 47 codes*)		
1. Guidance on Retrospection	**9**	**305**
Ask the *Performer* to recall what they saw	9	35
Ask the *Performer* to recall what they heard	8	14
Ask the *Performer* to recall what they did in specific situations	9	36
Ask the *Performer*'s thoughts during the experience	9	30
Ask the *Observer* to recall what they see	6	10
Ask the *Observers*' observation of the *Performer*'s action	3	7
Ask *Observers*' actions during the *Performer*'s VR simulation	2	2
Ask the *Observer*'s thoughts on the experience	5	11
Ask the group to recall detailed information	6	7
Ask the *Observer* to recall detailed information	6	14
Ask the *Performer* to recall detailed information	9	56
Lead further retrospection with critical cues	7	12
Guide on retrospection by describing the VR scenario	9	31
Guide on retrospection by describing the *Performer*'s action	6	19
Guide on retrospection by describing the group's action	2	3
Repeat the *Observer*'s description of their observation	3	6
Repeat the *Performer*'s description of the VR experience	8	12
2. Facilitation of Analysis and Explanation of Experience	**9**	**153**
Ask the group to interpret a critical sign or signal	9	29
Ask the *Performer* to interpret a critical sign or signal	5	8
Ask the *Observer* to interpret a critical sign or signal	6	9
Ask the group's opinion on hypothetical questions	9	24
Ask the *Performer* to answer hypothetical questions	8	10
Ask the *Observer* to answer hypothetical questions	4	6
Ask for decisions in a specific situation	9	42
Ask the *Performer* to explain certain behavior in VR	7	13
Ask the *Performer* to further elaborate on certain decision-making	3	5
Ask the *Observer* to further elaborate on certain decision-making	4	7
3. Support on Knowledge construction and Transfer	**9**	**104**
Ask about the resource of mentioned knowledge	1	2
Ask about daily behaviors beyond the VR experience	3	4
Clarify unclear/incomplete descriptions mentioned in the group	2	2

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Category|Open code, frequency	Frequency	
	Case	Code
Confirm the group's prediction in specific situations	2	3
Guide on safe pedestrian behaviors	9	51
Guide on pedestrian knowledge	9	32
Explain the reason for certain safe pedestrian behaviors	2	2
Summarize the safe pedestrian behavior or solution	6	8
4. Facilitation on Teamwork	**9**	**131**
Diagnose reasons for possible issues	5	8
Direct attention to a certain topic	9	25
Direct attention to those who are interrupted by others	3	4
Induce discussion by asking for agreement on certain opinions	2	3
Induce discussion by asking others' thoughts	9	46
Solicit all opinions in the group	2	3
Invite *Observers* to participate in the discussion	8	23
Invite *Performer* to participate in the discussion	3	8
Induce discussion by contrasting different opinions in the group	4	5
Induce discussion by interpreting the *Performer*'s action in VR	2	2
Induce discussion by suggesting ideas not mentioned by the group	1	1
Explain possible reasons for issues in a given situation	2	3
Role: *Performer* (*4 categories, 32 codes*)		
1. Retrospection and Description of the Experience	**9**	**211**
Describe what they heard (by answering questions)	6	11
Describe what they saw	9	39
Simply indicate what they saw by answering questions	1	6
Describe what happens in the VR scenario	5	15
Describe what happens to them in VR	2	2
Describe what they did in VR	8	30
Indicate what they did by simply answering questions	3	8
Exhibit what they did with body language/gestures	2	4
Describe their thoughts in specific situations	9	22
Can't illustrate their thought during VR (respond with silence or not know)	2	3
Indicate that they thought about nothing during VR	1	4
Point to a location on the map to indicate detailed information about their VR experience	6	19
Indicate heir confusion in a specific situation in VR	2	3
Forget detailed information in VR (cognitive load)	2	6
Miss critical cues in VR	3	6
Notice critical cues in VR	9	33
2. Analysis and Thinking about the Experience	**8**	**25**
Explain their actions in VR	4	10
Interpret a critical sign (by answering questions)	5	9
Interpret a critical sign with further explanation	4	4
Share heir daily pedestrian practice in a specific situation	1	2
3. Knowledge Construction and Decision Making	**9**	**41**
Make their decision in a given situation	9	21
Elaborate their decision-making by answering questions	1	1
Explain their decision-making in a given situation	2	2
Predict results from hypothetical behaviors	9	17
4. Interactive and Collaborative Meaning-making	**5**	**16**
Agree with others' decision-making in a given situation	2	2
Suggest alternative decision-making in a given situation	1	1
Complement others' opinions	1	1
Complement safe decision-making in a given situation	1	3
Complement the knowledge discussed in the group	1	2
Agree with others' prediction from hypothetical behaviors	3	3
Try to answer about their VR experience but is interrupted	2	3
Being upset when *Observers*' comment on their actions	1	1
Role: *Observer* (*4 categories, 54 codes*)		
1. Retrospection and Description of the Experience	**9**	**194**
Describe what they saw on the screen	9	33
Simply indicates what they saw by answering questions	5	7
Indicate what they saw beside the *Performer*'s description	3	6
Describe what they heard from VR	2	2
Describe the *Performer*'s actions in VR	7	22
Describe situational information related to the *Performer*'s action	6	13
Describe details that the *Performer* didn't remember	2	2
Mimic the *Performer*'s actions in VR to indicate what they see	1	3
Describe their actions during the *Performer*'s VR simulation	4	5
Describe their thoughts during the *Performer*'s VR simulation	4	4
Answer they thought nothing during the *Performer*'s VR simulation	5	6

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Category|Open code, frequency	Frequency	
	Case	Code
Describe their related experiences (for those who had VR experience)	1	3
Point to a location on the map to indicate their observation in VR	6	27
Comment on another *Observer*'s thoughts on the *Performer*'s actions	1	2
Comment on the *Performer*'s actions in VR	3	7
Have limited VR experience with audio modality	4	7
Answer questions regarding the VR events on behalf of the *Performer*	3	10
Miss critical cues in VR	3	6
Notice critical cues in VR	9	19
Notice the *Performer*'s actions in VR	6	10
2. Analysis and Thinking about the Experience	**9**	**80**
Interpret a critical sign (by answering questions)	9	28
Interpret the *Performer*'s actions in VR	3	5
Analyze safe pedestrian decisions in a specific situation	2	3
Can't explain their decision-making (respond with silence or do not know)	2	2
Try to explain their decision-making but is interrupted	1	1
Can't interpret a critical sign (by answering questions)	2	2
Explain their decision-making in a given situation	5	10
Predict results from hypothetical actions	9	29
3. Knowledge Construction and Decision Making	**9**	**64**
Ask questions related to their decision-making in a given situation	1	1
Complement others' interpretation of a critical sign or signal	2	5
Complement the knowledge discussed in the instruction stage	2	5
Complement the safe behavior discussed in the instruction stage	3	4
Make one's decision in a given situation	9	39
Plan for one's own VR experience later	1	3
Share one's daily pedestrian practice regarding a specific situation	3	3
Suggest alternative actions in a given situation	2	3
Summarize safe pedestrian behaviors in the discussion	1	1
4. Interactive and Collaborative Meaning-making	**9**	**67**
Agree with others' decision-making in a given situation	8	15
Agree with others' interpretation of a critical sign or signal	5	7
Agree with others' prediction from hypothetical behaviors	5	6
Agree with the *Performer*'s description of the VR scenario	3	5
Challenge others' opinion by showing disagreement	2	3
Challenge others' opinion by predicting potential results	2	2
Challenge others' interpretation by showing disagreement	2	3
Challenge others' decision-making by suggesting alternative situations	2	3
Challenge the instruction by suggesting a hypothetical situation	1	2
Challenge the *Performer*'s retrospection	1	1
Comment on others' decision-making in a specific situation	1	1
Complement on others' analysis	4	7
Deny their previous suggestion after getting challenged	1	1
Interrupt the *Performer* to explain the *Performer*'s actions in VR	1	2
Interrupt the *Performer* to describe their own experience instead	3	3
Interrupt the *Performer*'s retrospection and describes what they saw	2	3
Interrupt others to explain a certain decision making	2	2
The Group as A Unit (15 codes)		
The group complements the safe behaviors suggested in specific situations	2	2
The group interprets a critical sign by answering questions	6	8
The group interprets a critical sign by linking to a safety formula	1	1
The group predicts the same results from hypothetical behaviors	9	13
The group predicts different results from hypothetical behaviors	2	3
The group shares their daily pedestrian practice by answering questions	2	7
The group suggests the same decisions regarding a specific situation	5	11
The group suggests different decisions regarding a specific situation	3	6
The group agrees with the safe pedestrian behaviors suggested by the *Facilitator*	9	17
Observers' different focuses in observation	2	3
Observers' inconsistent experiences in catching cues in VR	1	1
Performer and *Observers*' inconsistent experiences in catching cues in VR	2	2
The group discusses the VR program (not related to the reflecting activity)	2	5
The group's off-task attention	3	10
The group's off-task interactions	2	4

Data availability

Data will be made available on request.

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